

This assignment contributes 10% towards your final grade. It should be submitted by 3pm on Friday 27 March 2026. In the absence of an approved extension, assignments submitted after the due date will attract a penalty as outlined in the course profile. Prepare your assignment as a **pdf file**, either by typing it, writing on a tablet or by scanning/photographing your handwritten work. Ensure that your name appears on the first page of your submission. Check that your pdf file is legible and that the file size is not excessive. Files that are poorly scanned and/or illegible may not be marked. Use of generative AI is permitted, but not encouraged for this assessment. If you choose to use AI, you must do so appropriately and provide clear acknowledgement and referencing in line with UQ's guidelines.

1. (8 marks) Assume you are working with the English alphabet.
 - (a) Encrypt the word **koala** using the “opposite” affine cipher $x \mapsto 25 - x$.
 - (b) You discover that the encryption function for an affine cipher with parameters a, b sends **help** to **NPTZ**. Can you determine a, b ? If yes, then do so, and if no, then explain why not.
 - (c) Show the choice of parameters in point (b) is not a good one, by finding two different plaintexts that encrypt to the same ciphertext.
2. (10 marks) Consider the Hill cipher defined by the following matrix:

$$M = \begin{pmatrix} 14 & 13 & 0 \\ 13 & 14 & 0 \\ 13 & 13 & 1 \end{pmatrix}$$

- (a) Use M to encrypt the message **bobcat**.
 - (b) Use M to encrypt the ciphertext you obtained in point (a). What do you notice?
 - (c) What can you deduce about the value of the determinant of M , without performing computations?
3. (8 marks)

Your chemist friend has a rival, who has encrypted the name of the required reagent for their new process. You find out it is one of: **carbon, oxygen, sodium, copper, silver**. You also discover they encrypted it as **NQDETV** using an autokey cipher. Can you help your friend to find out the correct reagent?
4. (12 marks) You will be emailed some ciphertext that was encrypted from an English plaintext using a Vigenère cipher. You are encouraged to write (and submit) your own computer code to answer this question, appropriately documented so as to demonstrate your understanding. If you prefer not to write code, you are welcome to use readily available online tools to answer these questions, in which case you must reference the tools that you use, and write a brief explanation for each question part to demonstrate that you understand what the tools are doing.
 - (a) Use Friedman's method 2 to estimate the length of the key that was used to encrypt your given ciphertext.

- (b) Use Kasiski's test or Friedman's method 1 to determine the most likely length of the key that was used to encrypt your given ciphertext.
- (c) Decrypt your given ciphertext and state the keyword that was used to encrypt it. (You don't need to write out the entire plaintext, just the first sentence is sufficient.)

5. (10 marks) Find all integer values of a for which the following system has a solution for $x \in \mathbb{Z}$, and then solve it:

$$\begin{cases} 3^{x^2-1} \equiv 2^a & \text{mod } 13 \\ x - 1 \equiv 0 & \text{mod } 3. \end{cases}$$

6. (12 marks) Alice decides to use RSA to allow Bob to send her messages, and makes

$$(N, e) = (63187, 13)$$

her public key. Through a breach in security, you manage to find out that $\phi(N) = 2^6 \cdot 11 \cdot 89$, and to intercept Bob's very important numerical message $C = 50521$.

- (a) Determine p, q .
- (b) Determine Alice's private key.
- (c) Decrypt C , if possible.